

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – MCQ Test

|                  |                                                                                                                                                 |               |                              |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------|
| Course Code:     | CSA0309                                                                                                                                         | Course Title: | Data Structures              |
| CO Assessed:     | CO3 – Precision (BL3, BL5)                                                                                                                      | Assessment:   | Assessment Tool 3 – MCQ Test |
| Weightage:       | 30% of CIA                                                                                                                                      | Total Marks:  | 100                          |
| CO3 Statement:   | <i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.</i> |               |                              |
| Student Name:    | M.Mary Sharlin                                                                                                                                  | Reg. No.:     | 192511330                    |
| Section / Batch: | BE CSE                                                                                                                                          | Date:         | 10/08/2026                   |

### Code of Conduct

I M.Mary Sharlin ( 192511330 ) (Name / Reg No)  
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.Mary Sharlin

### Instructions

- This test contains 25 MCQ Test questions (4 marks each), Total: 100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements:  
Identify the most appropriate answer, conceptual understanding, problem-solving ability, application of knowledge.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic                  | Focus                                                                  | Q Nos | Marks     |
|----------------------------------|------------------------------------------------------------------------|-------|-----------|
| Preliminaries – Tree Terminology | Root, height, level, siblings and sub tree                             | 1–2   | 8         |
| Binary Tree, Binary Search Tree  | Properties, binary tree and binary search tree, insertion and deletion | 3–9   | 28        |
| Tree Traversals                  | Traversal types, In order, Post order and Pre order                    | 10–14 | 20        |
| AVL Tree                         | Balancing factor, Rotation, Types of rotation, examples                | 15–16 | 8         |
| Applications of Search Trees     | Usage of binary tree                                                   | 17    | 4         |
| B Tree - 2-3 tree, 2-3-4 tree    | Properties, Example and Operations                                     | 18–19 | 8         |
| TRIE                             | Properties, structures and Operations                                  | 20    | 4         |
| Red Black Tree                   | Properties, Balancing and Operations                                   | 21–22 | 8         |
| Splay Tree                       | Splaying, Types and examples for insertion and deletion                | 23–25 | 12        |
| GRAND TOTAL:                     |                                                                        |       | 100 Marks |

## MCQ

### Rubrics for Calculation

| Criteria                 | Weightage | Level 4 – Exemplary                                                                                                 | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                                    |
|--------------------------|-----------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| Conceptual Understanding | 20        | Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge | Shows good understanding with minor conceptual gaps         | Basic understanding; several misconceptions present | Limited understanding; major concepts misunderstood    |
| Application Skills       | 20        | Correctly applies concepts to solve complex and scenario-based MCQs                                                 | Applies concepts correctly in most moderate-level questions | Struggles with application in complex scenarios     | Unable to apply concepts effectively                   |
| Analytical Thinking      | 30        | Consistently selects correct answers for high-order questions                                                       | Performs well in moderate analytical questions              | Limited success in analytical questions             | Unable to interpret analytical/problem-based questions |
| Time Management          | 20        | Completes test efficiently with time for review                                                                     | Completes test within time                                  | Struggles to complete test on time                  | Unable to complete test                                |
| Accuracy of Responses    | 10        | 90–100% correct answers                                                                                             | 75–89% correct answers                                      | 50–74% correct answers                              | Below 50% correct answers                              |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

**Q1 Tree Terminology**

4 marks

Nodes having the same parent are called:

A. Ancestors

B. Descendants

C. Siblings

D. Leaves

 Save Answer

Next →

**Q2 Quad Tree**

4 marks

In a Quad Tree, each internal node has at most:

A. 2 children

B. 3 children

C. 4 children

D. 8 children

 Save Answer

Next →

**Q3 Splay Tree**

4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf

B. Middle

C. Root

D. NULL

 Save Answer

Next →

**Q4 Splay Trees**

4 marks

Identify the rotation:

```
50
 \
 70
 \
 90
```

**A.** Zag-Zag

**B.** Zig-Zig

**C.** Zig

**D.** AVL Rotation

 Save Answer

Next →

**Q5 Splay Tree**

4 marks

Splay Tree is a type of:

**A.** Complete Binary Tree

**B.** Self-adjusting Binary Search Tree

**C.** Heap Tree

**D.** Graph

 Save Answer

Next →

**Q6 Red-Black Tree**

4 marks

If a red node has a red child after insertion, what is this called?

**A.** Black-height problem

**B.** BST violation

**C.** Root violation

**D.** Red-Red violation

 Save Answer

Next →

**Q7 Red-Black Tree**

4 marks

Insert sequence 70, 50, 90, 40 causes insertion of 40 under red parent. Uncle is red. What is needed?

A. Rotation

B. Recoloring

C. Deletion

D. No change

 Save Answer

Next →

**Q8 Red-Black Tree**

4 marks

If black height differs in two paths from the same node, the tree becomes:

A. Valid

B. Invalid

C. Balanced

D. Complete

 Save Answer

Next →

**Q9 Red-Black Tree**

4 marks

Every path from a node to its descendant NULL nodes must have the same number of

A. Black nodes

B. Red nodes

C. Leaf nodes

D. Parent nodes

 Save Answer

Next →

Q10 2-3 Tree

4 marks

In a balanced 2-3 Tree, all leaves must be at:

A. Different heights

B. Root level

C. Random height

D. Same height

 Save Answer

Next →

Q11 2-3-4 Tree

4 marks

Insert 10, 20, 30, 40 in a 2-3-4 Tree. What happens after inserting 40?

A. Tree becomes unbalanced

B. Node split may occur

C. Node is deleted

D. No change

 Save Answer

Next →

Q12 2-3 Tree

4 marks

The maximum number of children in a 2-3 Tree node is:

A. 3

B. 2

C. 5

D. 4

 Save Answer

Next →

**Q13 TRIE**

4 marks

In Trie, 'm' represents:

A. Number of nodes

B. Length of the key

C. Number of edges

D. Height of tree

 Save Answer

Next →

**Q14 B-Tree**

4 marks

In a B-Tree of order 5, the maximum number of keys in a node is:

A. 4

B. 5

C. 6

D. 3

 Save Answer

Next →

**Q15 B-Tree**

4 marks

What is the minimum number of children for a non-root node in a B-Tree of order 'm'?

A.  $m/2$

B.  $m-1$

C.  $m$

D.  $\lceil m/2 \rceil$

 Save Answer

Next →

**Q16 Search Tree**

4 marks

The main goal of using search trees is:

- A. Increase complexity
- B. Reduce storage to zero
- C. Improve efficiency of data operations
- D. Avoid using memory

 Save Answer

Next →

**Q17 AVL Trees**

4 marks

AVL trees are preferred over ordinary BST because they:

- A. Use less memory always
- B. Maintain balance for faster operations
- C. Do not require insertion
- D. Have no root node

 Save Answer

Next →

**Q18 Binary Search Trees**

4 marks

Which of the following is a common application of Binary Search Trees (BST)?

- A. Database indexing
- B. Video editing
- C. Audio compression
- D. Network wiring

 Save Answer

Next →



**Q19 AVL Tree**

4 marks

Consider the AVL tree:

30

/

10

\

20

Which rotation is required?

A. Left Rotation

B. Right Rotation

C. Left Rotation + Right Rotation

D. Right Rotation + Left Rotation

**Q20 AVL Tree**

4 marks

The maximum height difference allowed between left and right subtrees is:

B. 1

C. 2

D. 3

 Save Answer

Next →

**Q21 Binary Search Tree**

4 marks

A data compression system uses a tree-based model to assign shorter codes to frequently used patterns. You need a self-adjusting BST where frequency is not stored explicitly. Which is most appropriate?

A. AVL Tree

B. Threaded Binary Tree

C. B-Tree

D. Splay Tree

 Save Answer

Next →

**Q22 Tree Traversal**

4 marks

Consider the binary tree:

A

/ \

B C

/ \ \

D E F

What is the Preorder traversal?

A. D B E A C F

B. A B D E C F

C. D E B F C A

D. B D E A C F

**Q23 Tree Traversal**

4 marks

You are debugging a program that reconstructs the original root of a binary decision tree using traversal logs: Inorder and postorder. Given: Inorder: D B E A F C Postorder: D E B F C A What is the root?

A. A

B. B

C. C

D. D

 Save Answer

Next →

**Q24 Binary Tree**

4 marks

The maximum number of nodes in a binary tree of height  $h$  is:

A.  $h$

B.  $2^h - 1$

C.  $2^{(h+1)} - 1$

D.  $h^2$

 Save Answer

Next →

**Q25 Tree Terminology**

4 marks

A path from a node to the root is formed by its:

A. Descendants

B. Ancestors

C. Children

D. Leaves

 Save Answer